

Young Hyun Kim

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RESEARCH INTERESTS

Human cardiac tissue engineering and organoid models, chamber-specific electromechanical function, mechanobiology, and quantitative phenotyping of hiPSC-derived cardiac systems.

EDUCATION

Sejong University Seoul, Republic of Korea

M.S. in Integrative Biological Sciences and Industry Aug 2026

Thesis: *Generation of Chamber-Specific 3D Cardiac Organoids and an Integrated Electro-Mechanical Optical Analysis Platform.*

B.S. in Integrative Biological Sciences and Industry Aug 2023

RESEARCH EXPERIENCE

Biological Systems Engineering Laboratory, Sejong University Seoul, Korea

Dept. of Integrative Biological Sciences and Industry · PI: Prof. Keel Yong Lee

Research Assistant

Sep 2026 – Present

Graduate Research Assistant

Sep 2023 – Aug 2026

Thesis Project: Chamber-Specific Cardiac Organoid Platform

- Engineered chamber-specific atrial- and ventricular-like 3D cardiac organoids via temporal Wnt/retinoic acid modulation.
- Characterized chamber identity by qRT-PCR, immunofluorescence, and flow cytometry, and designed a MATLAB-based morphology analyzer validated against manual measurements.
- Developed and validated iCAMAnalyzer to integrate calcium kinetics, activation/conduction mapping, and label-free contractility in a single Python workflow; applied it to chamber-specific cardiac organoids and validated analysis performance using synthetic ground-truth videos.

Additional Research & Collaborations

- Optimizing a spinner-flask protocol to scale hiPSC-cardiomyocyte production for higher-throughput organoid fabrication.
- Differentiated and supplied hiPSC-derived cardiomyocytes and jointly performed flow cytometry characterization for a co-authored study in *Cancer Research Communications*, contributing to Resources and Visualization.

Undergraduate Research Assistant

Oct 2022 – Aug 2023

- Developed the laboratory's initial hiPSC-derived cardiac organoid fabrication protocol, establishing a foundation for subsequent organoid research.
- Fabricated C2C12-based 3D skeletal muscle tissues and applied cyclic mechanical stretch to assess tissue maturation.
- Founding laboratory member: As one of three core members, contributed to establishing the laboratory's initial infrastructure, including equipment setup, core experimental SOPs and safety procedures.

CONFERENCE PRESENTATIONS

- Poster, Korea BioChip Society (Nov 2025, Jeju) — Cardiac Organoid Platform with Micropillar-Assisted Calcium and Optical Signal Analysis
- Poster, 2025 Summer Symposium on Biological Convergence Engineering (Sep 2025, Sejong University) — Chamber-Specific Cardiac Organoid Platform with Micropillar-Assisted Calcium and Optical Signal Analysis

PUBLICATIONS

1. **Kim, Y. H.**, Son, Y. H., Choi, Y., Kim, M. S., Park, S. J., & Lee, K. Y. (2026). Advanced in vitro cardiac models for drug evaluation: Integration of organoids, engineered tissues, and microphysiological systems. *Microsystems & Nanoengineering*, 12(1), 162. **(Review)**
2. Kim, M. S., **Kim, Y. H.**, Choi, Y., & Lee, K. Y. (2025). Biohybrid actuators in robotics: recent trends and future perspectives of skeletal and cardiac muscle integration. *npj Robotics*, 3(1), 37. **(Review)**
3. Kim, Y. S., Lee, M., Cheon, Y. J., Choi, H. S., Choi, B. J., **Kim, Y. H.**, Lee, K. Y., & Ryu, C. J. (2026). Targeting CD147 or CD98hc is sufficient to collapse the CD147–CD98hc–MCT1/MCT4 module and restrain hepatocellular carcinoma growth. *Cancer Research Communications*. **Accepted for publication.**

INSTRUMENTATION & ENGINEERING PROJECTS

- **Bio-Synchronous 3D Tissue Electromechanical Stimulator (2025–present):** Contributing to the functional redesign of an existing stretch system to integrate mechanical stretch, electrical stimulation, and controlled culture conditions; defining experimental requirements and coordinating implementation with an external fabricator.
- **Dual-Camera Fluorescence/Brightfield Imaging System (2026–present):** Developing a custom dual-camera optical system to enable simultaneous measurement of calcium dynamics and tissue motion in engineered cardiac tissues; integrating dichroic wavelength separation, fluorescence/brightfield channels, and camera/filter components on an Olympus SZX7 microscope.

TEACHING AND MENTORING — Sejong University

Undergraduate Research Mentor — SURF Program

Jul 2026 – Aug 2026

Mentored 4 undergraduate researchers in laboratory onboarding, C2C12 and hiPSC culture, experimental planning, confocal imaging, and preparation of final project presentations.

Teaching Assistant — Biochemistry Laboratory

Jul 2023 – Dec 2024

Guided ~120 students through antibody-production workflows including plasmid transformation, protein expression and purification, and Western blotting; prepared instructional slides and laboratory materials.

Teaching Assistant — Microbiology Laboratory

Jan 2024 – Jun 2024

Instructed 40 students in bacterial culture, Gram staining, morphological characterization, biochemical testing, and API-based bacterial identification.

Peer Research Mentor — DoLearn-DoRun Program

Sep 2023 – Dec 2023

Mentored 2 undergraduate researchers on an in vitro muscle model using line-patterned substrates to promote myotube alignment; authored the project proposal and directed experimental work.

SKILLS

Cell & Tissue Engineering: hiPSC culture and chamber-specific cardiomyocyte differentiation; 3D cardiac organoid and skeletal muscle tissue engineering.

Assays & Imaging: flow cytometry, qRT-PCR, immunofluorescence/confocal microscopy, calcium-transient imaging, and contractility analysis.

Computational Analysis & Programming: MATLAB (App Designer), Python (Streamlit, NumPy/SciPy), ImageJ, GraphPad Prism; custom analysis-tool development.

Simulation & Fabrication: COMSOL Multiphysics; Rhino/3D modeling; 3D printing.

Scientific Illustration: Inkscape, Adobe Illustrator.